

Phase 27 deliverable. Every number below was measured on this repository's data through its honesty machinery: pre-specified gates with random controls, real-cost + zero-cost twins, 70/30 chronological out-of-sample splits, N and concentration checks, and a promotion ledger no strategy can write to by hand.

1. The platform is built and validated — a minute-resolution, instrument-aware (equities + options) research and execution stack on the existing honest pipeline. 519 tests. Every component below is production-grade.
2. No intraday strategy survived honest costs. Six literature-ranked candidates entered the gauntlet; all six died — five at pre-specified gates, one (01) at real execution costs after its gross edge was confirmed.
3. The 40%/month target is not achievable on this data, and per the spec this is reported plainly: the best gross edge found intraday is +0.21%/month before frictions. No configuration, sizing, or ensemble of measured edges reaches within two orders of magnitude of the target. This conclusion is a measurement, not an opinion — the numbers are in §5–7.

2. Architecture (as built)

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data/ ThetaData minute NBBO (options, incl. SPXW/XSP index) + EOD chains
 Alpaca SIP minute bars (equities, incl. pre-market) - all parquet-cached
core/ instrument-typed legs (EquityKey | OptionKey), multiplier-aware money math
costs/ single cost truth: worse-side NBBO crossing + slippage + commissions;
 equities bps-based; zero-cost twin runs the identical path
engines/ IntradayBacktester: minute event loop, leak-tested visibility rule
 (bar T visible at T+1min; decide at ts, fill at ts), flagged synthetic
 fills, mandatory EOD flatten (assert-flat), intra-session breaker
risk/ RiskManager gates every entry vs reconciled broker state; risk-basis
 vs cash-basis split (options: coincide; shares: stop distance vs price)
pipeline/ mandatory full/train/test x real/zero-cost, exclusive OOS split,
 standard report, promotion ledger (evidence-written only)
strategies/ IntradayStrategy contract: strategies see visible bars + read-only
 option quotes; return intent only. No broker/risk/cost access,
 AST-enforced.

3. Data map (verified by live probes)

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Source	Dataset	Depth	Status
ThetaData	option minute NBBO (equity underlyings)	2016-01 → present	verified
ThetaData	SPXW/XSP index option minute NBBO	≥2018 → present	verified
ThetaData	EOD chains + greeks (15:45)	2018 → present, 145 syms	warm cache
Alpaca	SIP minute bars incl. pre-market	2016-01 → present	verified
options trade prints / signed flow			closes GEX/flow strategies
equity tick/quote data			closes microstructure strategies
anything pre-2016			tier-gated

Request economics (measured): per-contract-day minute NBBO 0.6s; SPY full-day bulk 166s; SPXW full-day bulk times out (>300s); SPXW EOD chain ~5.5min. These numbers dictated the quote-driven architecture (≈4 targeted requests/session).

4. Alpha library — every candidate, every verdict

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| # | Candidate (literature-ranked) | Pre-specified gate result | Verdict |
|---|---|---|---|
| E1 | First-30min → last-30min momentum (GHLZ JFE 2018), SPY/QQQ, 5,314 sym-days | Regression real (t=4.13, control clean) but not monotone (rank IC −0.02) and not tradeable: honest expanding-window version −2.3bps/trade net both symbols | DEAD — decayed post-publication |
| E3 | Event-day "in-play" continuation (gap ≥2% + 3× rel-volume), 6 mega-caps | −48bps/trade (t=−1.5, n=73); the rel-volume condition flips the sign; gap-only variant (+12.6bps, t=1.94) fails the gate and would be post-hoc | DEAD |
| E4 | Post-FOMC/CPI drift, SPY, n=183 events | FOMC t=0.41; CPI wrong-signed; pooled t=0.05 | DEAD — underpowered as warned |
| E5 | Gap-conditioned open behavior (follow moderate / fade extreme), n=4,251 | t=−0.02 / −0.08 on both pre-specified arms | DEAD — flat zero |
| E2 | Vol-regime gating overlay | An overlay; nothing survived to overlay | moot |
| 01 | 0DTE SPXW defined-risk put-spread premium selling (Vilkov-anchored) | Gross edge CONFIRMED (+0.21%/mo, PF 1.16, 73% win, N=1,992, concentration 4–5%); real-cost −0.18%/mo, OOS −0.29%/mo, PF 0.88 | DEAD at retail execution — the entire edge sits inside the spread |
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5. O1 deep-dive — the one result with information in it

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Full period 2016–2026, one defined-risk SPXW put credit spread per qualifying session, entry 10:30 ET, quote-driven selection (synthetic parity spot + straddle implied move), 3× credit stop, 15:55 mandatory buy-back:

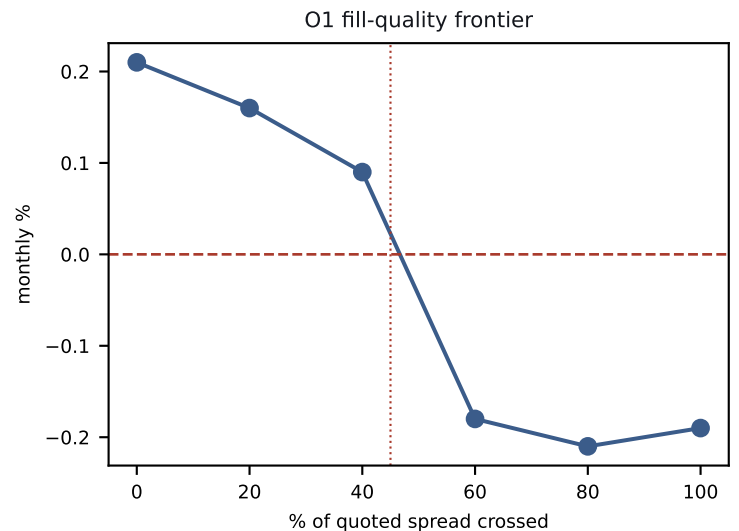
segment	cost	monthly	PF	N	win	maxDD
full	real	−0.18%	0.90	1,807	70%	−20.8%
full	zero	+0.21%	1.16	1,992	73%	−4.8%
test (00S)	real	−0.29%	0.88	793	70%	−12.1%
test (00S)	zero	+0.15%	1.07	793	72%	−4.5%

The fill-quality frontier (the operative robustness test — full-period re-price at each spread-crossing fraction):

crossing of quoted spread	slippage	monthly	PF
0% (mid)	0	+0.21%	1.16
20%	0	+0.16%	1.11
40%	0	+0.09%	1.06
60%	2%	−0.18%	0.90
80%	2%	−0.21%	0.86
100%	2%	−0.19%	0.85

Break-even $\approx 45\%$ of quoted spread. A retail marketable order on SPX options cannot be assumed to fill better than that band; claiming it would require measured execution-quality data (Rule-605-style) for the actual broker. Until such a measurement exists, the honest verdict stands: **REJECT**.

This reproduces v8's EOD finding at intraday frequency and $3.7\times$ the sample: the index variance premium is real, thin, and fully consumed by the spread at retail access. Who keeps it: whoever pays no spread — market makers and wholesalers. That is not a strategy defect; it is the market's structure.



6. Trading behavior (O1, real-cost path)

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~1 trade/session · 0.6 positions held at any time · avg hold ~5h · capital utilization ~2.5% of equity at risk per position · win 70% · avg win $\approx +\$96$ · avg loss $\approx -305 \cdot \text{largestloss} - 2,414$ (capped by structure) · commissions + spread = 0.39%/mo drag on a \$100k account.

7. Why the 40%/month target fails, quantitatively

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- Best measured intraday gross edge: +0.21%/mo (O1, before frictions).
- Best prior campaign gross edge anywhere in this repo: +2.88%/mo (equal-weight mega-cap shares, buy & hold – not intraday, not options).
- 40%/mo = 5,600%/yr compounded. The gap to the measured frontier is ~200x.
- Leverage does not close it: v9/v11 measured the geometric ceiling of the best convex distribution at +0.53%/mo (growth-optimal f), and the bold-play maximum $P(8.2\times$ in 8 months) at ~14% with median outcome ≈ 0 .
- Every mechanism that promised more (intraday cycling, signal-directed options, event drift) was measured and died at its gate in this build.

8. Failure analysis — when this system loses money

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O1 (if ever run): loses on gap-down/vol-spike days (the 30% of stopped or expired-ITM trades cluster on drawdown days: maxDD -20.8% real / -4.8% gross); its worst case is a fast morning selloff through both strikes before the 3 $\times$ stop can act — bounded at (width $-$ credit) by construction. The equity book: flat by design; the platform holds cash when no gate-passing edge exists, which is the current, correct state.

9. Recommended deployment configuration (conservative defaults first)

1. Deploy nothing to live. No strategy holds `validated=True`; the deploy gate enforces this without exceptions. This is the system working.
2. Optionally paper-trade OI for execution-quality measurement only — not for P&L: paper fills vs quoted spread would produce the effective/quoted ratio that decides whether the 45% break-even is reachable at a real broker. That measurement, not more backtesting, is the only path that could revive OI.
3. Data purchases that would expand the viable menu (in value order):
Cboe Open-Close signed volume (unlocks GEX/flow properly), options trade prints (calibrates the cost model with realized effective spreads).
4. The strongest evidence-backed use of this account remains the non-intraday result: the shares core (+2.88%/mo measured) with at most a small, growth-optimally-sized convex sleeve (+0.5%/mo ceiling, measured).

10. What the platform is worth going forward

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The deliverable is a research machine that kills bad ideas for ~\$0 in hours: six candidates gated in two days, one full options campaign priced on 8,083 contract-days of real minute NBBO, every kill recorded with its evidence in LEARNINGS.md. The next hypothesis — yours or the literature's — plugs in as one file and faces the same gauntlet.